Lecture 6

Tuesday, 1 September 2026 7:28 PM

Random Variables

$(\Omega, \mathcal{F}, \mathbb{P})$ – Prob. Space

- Ω → Sample space
- $\mathcal{F}$ → Collection of events
- $\mathbb{P}$ → Prob. measure

① Toss coin n times. No of heads.

② Roll two dice. Sum of the outcomes

A random variable is a fn $X : \Omega \to \mathbb{R}$

Eg ① n Coin tosses : $(\omega_1, \omega_2 \ldots \omega_n) \to \vec{\omega}$

$X(\vec{\omega})$ = # heads in the tuple

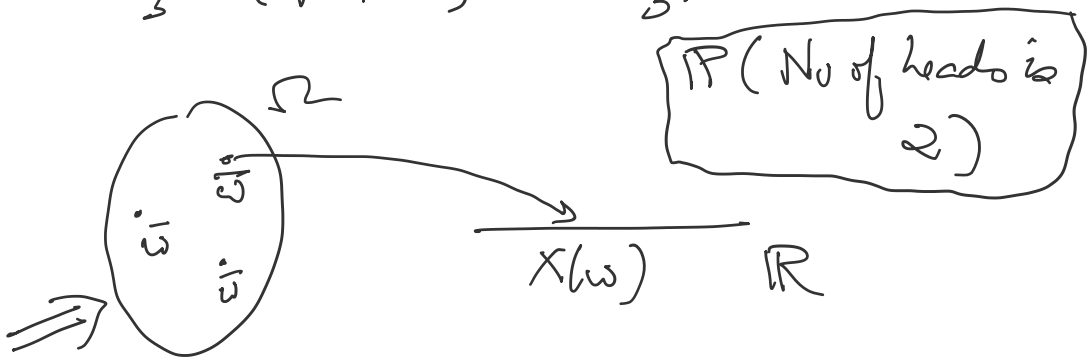
$n = 3$

$\vec{\omega}_1 = (H, T, H)$ $X(\vec{\omega}_1) = 2$

$\vec{\omega}_2 = (T, T, T)$, $X(\vec{\omega}_2) = 0$

$\vec{\omega}_3 = (T, H, T)$, $X(\vec{\omega}_3) = 1$

$\mathbb{P}$(No of heads is 2)



Random variable X is characterized by Cumulative Distribution Function (CDF)

$F_X : \mathbb{R} \to [0, 1]$

$F_X(x) \triangleq \mathbb{P}(X \le x)$

$\to \mathbb{P}(\{\omega : X(\omega) \le x\})$

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Cumulative Dist. Function (CDF)

2 Coin tosses $n=2$

$\Omega = \{HH, TH, TT, HT\}$

$\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$

$P(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{4}$

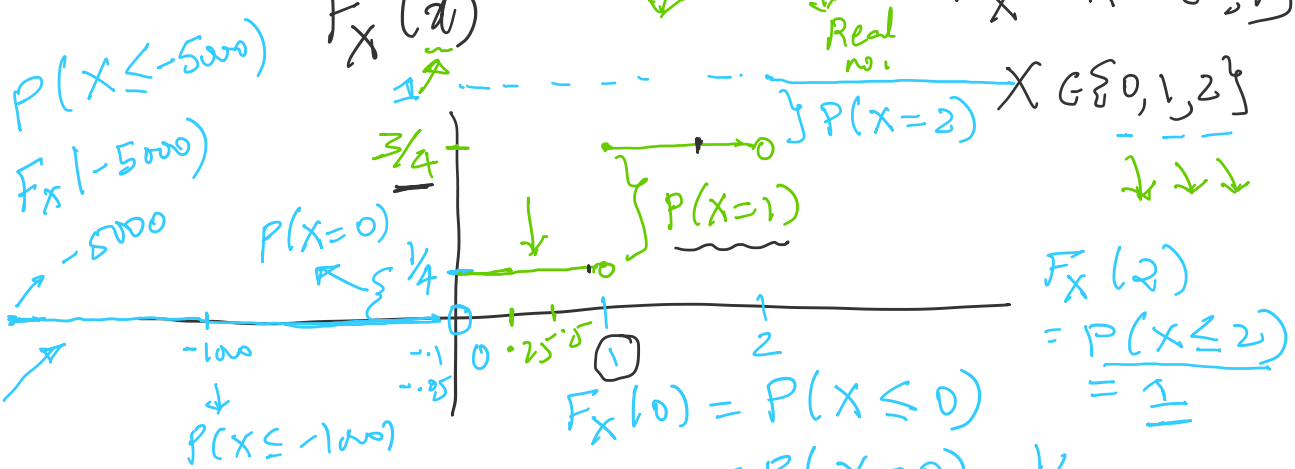
$P(A) = \sum_{i \in A} P(\{i\})$

X = # heads $\in \{0, 1, 2\}$

CDF $= P(X \le x)$

$F_X(x)$

$F_X : \mathbb{R} \to [0,1]$



$F_X(2) = P(X \le 2) = 1$

$F_X(0) = P(X \le 0) = P(X=0) = \frac{1}{4}$

$F_X(.25) = P(X \le .25) = P(X=0) = \frac{1}{4}$

$F_X(1) = P(X \le 1) = P(X=0) + P(X=1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$

$F_X : \mathbb{R} \to [0,1]$ $\quad F_X(x) \triangleq P(X \le x)$

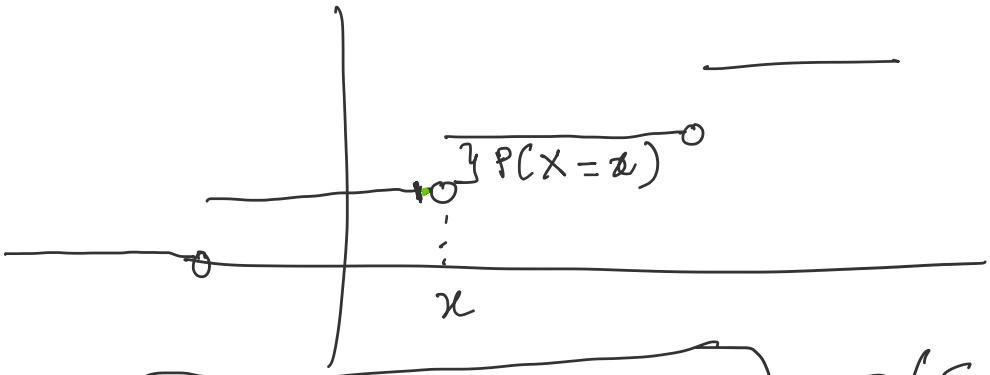
1. Monotonically non-decreasing
2. Jumps. Jump at $x \to P(X=x)$

$F_X(.25) = P(X \le .25) = P(X=0)$

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$F_X(0.75) = P(X=0)$

$F_X(0.99) = P(X=0)$



$F_X(x) \triangleq P(X \le x) = P(\{X<x\} \cup \{X=x\})$

$= P(X<x) + P(X=x)$

$\approx P(X \le x-\varepsilon) + P(X=x)$

$F_X(x^-)$

$F_X(x) = P(X \le x)$

$P(X=x) = F_X(x) - F_X(x^-)$

$P(a < X \le b)$

$P(X \le b) = P(\{X \le b\} \cap \{X \le a\}) + P(\{X \le b\} \cap \{X > a\})$

$P(X \le a) = F_X(a)$

$F_X(b)$

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$P(a \le X \le b)$

$$P(a \le X \le b) = F_X(b) - F_X(a)$$

$$P(a \le X \le b) = P(X < b) - P(X \le a)$$

$$= F_X(b^-) - F_X(a)$$

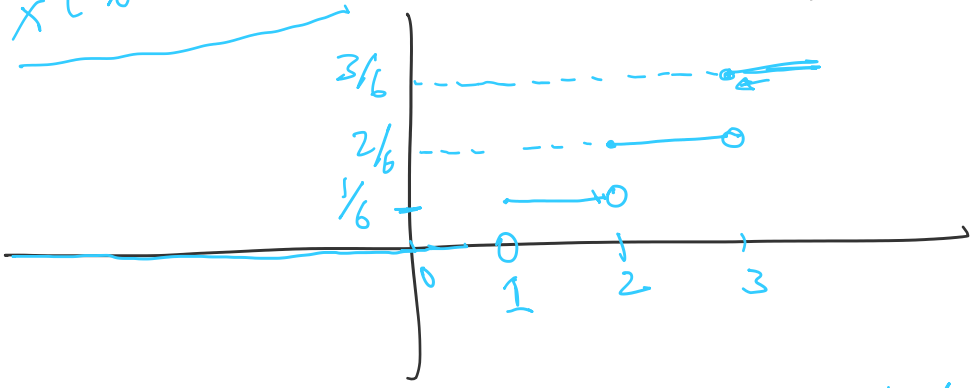
Roll a fair dice. X = outcome

$X \in \{1, 2, 3, 4, 5, 6\}$

$P(X = x)$

$P(X \le x)$

$F_X(x^+) = F_X(x)$



$$P(X = 2) = F_X(2) - F_X(2^-) = 1/6$$

$$P(1 < X \le 3) = F_X(3) - F_X(1)$$

$$= 3/6 - 1/6 = 2/6$$

$P(X = 2) + P(X = 3)$

$1/6 \quad 1/6$

$$P(2 \le X \le 5) = F_X(5^-) - F_X(2)$$

$$= P(X < 5) - P(X \le 2)$$

$$\downarrow$$

$$F_X(5^-) - F_X(2)$$

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$= \frac{4}{6} - \frac{2}{6} = \frac{2}{6}$

Types of RVs – (1) Discrete (2) Continuous

Discrete RVs → support is always Countable

Countable → finite / Countably infinite $\{1, 2, 3, \ldots\}$

X = # tosses need to see first head

$\{1, 2, 3, 4, \ldots\}$

Probability Mass function (PMF)

$$P_X(i) \triangleq P(X = i), \quad i \in S_X$$

Toss fair die → PMF X

$P(X = i) = \frac{1}{6}$ for $i \in \{1, 2, 3, 4, 5, 6\}$

Toss Coin. H, T

H → .2, T → .8

PMF X

$P(X = H) = .2$

$P(X = T) = .8$

PMF ⟷ CDF

$$P(X = i) = F_X(i) - F_X(i^-)$$

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$$F_X(x) = P(X \le x)$$

$$= \sum_{\substack{i \in S_X \\ i \le x}} P(X = i) = \sum_{\substack{i \in S_X \\ i \le x}} P_X(i)$$

Common Discrete RVs

(1) Bernoulli RV $\to \{0, 1\}$

$P(X=0) = 1-p$

$P(X=1) = p$

$\text{Ber}(p) \quad p \in (0, 1)$

(2) Binomial RV $- \text{Bin}(n, p) \to$ # success in n trials

$n \to$ # experiments

$p \to$ success prob. of each exp

$\in \{0, 1, 2, \ldots n\}$

$$\text{PMF}: P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$k \in \{0, 1, 2 \ldots n\}$

a toss, # heads $\to \text{Bin}(n, p)$

1000 emails, $P(\text{email opened}) = .1$

emails opened $\sim \text{Bin}(1000, .1)$

(3) Geometric RV $\to$ # tosses before seeing first heads

$P(\text{heads}) = p$

$$P(X = k) = (1-p)^{k-1} \cdot p \quad , k \in \{1, 2, 3 \ldots\}$$

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'Memoryless' → $P(X > s+t \mid X > s) = P(X > t)$

$s, t \in \mathbb{N}^+$

(4) Poisson (λ) ~ Poi (λ)

$\in \{0, 1, 2, 3 \ldots\}$

$P(X = k) = \dfrac{e^{-\lambda} \cdot \lambda^k}{!k}$ $\quad k \in \{0, 1, 2 \ldots\}$

Poisson → model traffic

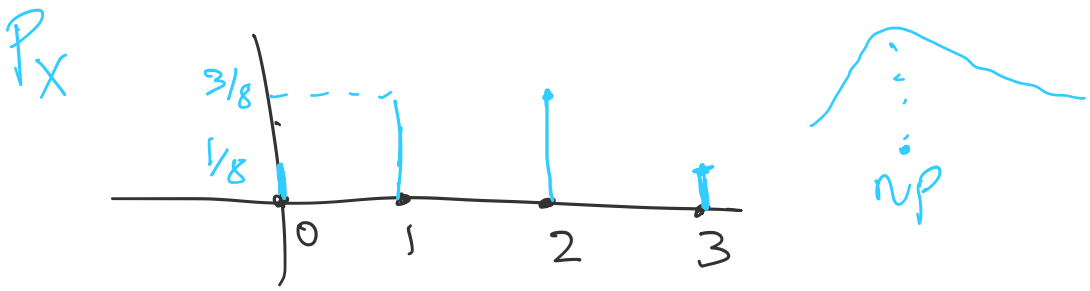
Bin (n, p) $\quad n = 3, \; p = \frac{1}{2}$

$X = 0 \rightarrow \binom{3}{0}\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$X = 1 \rightarrow \binom{3}{1}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^2 = 3 \times \frac{1}{8}$

$2 \rightarrow \binom{3}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right) \rightarrow \frac{3}{8}$

$3 \rightarrow \binom{3}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^0 \rightarrow \frac{1}{8}$



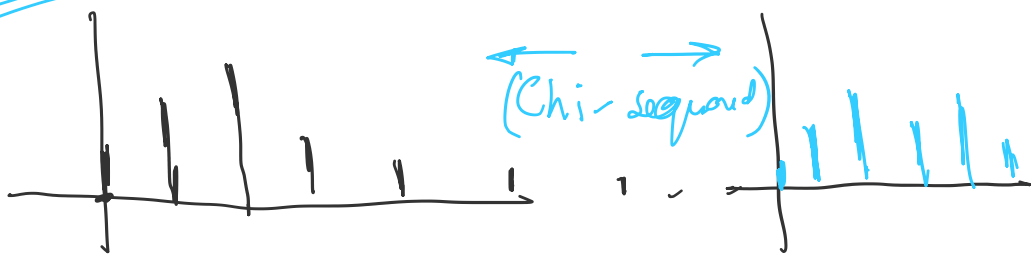
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Traffic arriving at my website b/w 9-10 AM

$Poi(\lambda)$ $\quad \lambda \rightarrow$ Fit my historical data

$\hat{\lambda}$ = Avg no of users / day

$Poi(\hat{\lambda})$



$$P(2 \leq X \leq 5)$$

$$= P(X \leq 5) - P(X < 2)$$

$$= F_X(5) - F_X(2^-)$$

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